



SUJET STATAPP 353 - 2026-2027



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Organisme : CREST

Catégorie : Environnement, Energie, Mobilité

Sujet :

When a critical material becomes scarce, do firms dig in and innovate around it, or pivot into new activities? And does their existing technological know-how pull them toward the same constrained input?

Description du sujet :

Why now?

Two of the defining industrial transformations of this decade are competing for the same limited pool of materials. The ecological transition runs on critical minerals. Electric-vehicle batteries depend on lithium, cobalt, nickel and graphite. Wind turbines and EV motors depend on rare-earth permanent magnets. Solar and grid infrastructure depend on silicon, silver and copper. Supply of many of these materials is geographically concentrated, and it has become a policy lever: since 2023, China has introduced export controls on gallium, germanium, graphite and several rare earths. These are supply shocks hitting exactly the industries that decarbonisation relies on.

The AI boom adds a demand shock on top. Chips, data centres and the power infrastructure behind them need semiconductor-grade materials, copper and energy-grid components. This pushes up demand for inputs that transition industries also need.

For firms, a material shock is more than a cost problem; it is a strategic question about which activities to keep, which to abandon and which to enter. The answer shapes how fast green technologies scale, how resilient supply chains become, and whether industrial policy aimed at "de-risking" actually changes what firms produce.

Research question and motivation:

How do material constraints shape firms' production diversification?

A disruption in material supply can make some activities less profitable and make entering others more appealing. But there is a catch. The activities closest to a firm's existing know-how, where diversification is cheapest, may depend on the very same constrained input. Firms therefore face a trade-off between technological proximity and material exposure.

We study how firms navigate this trade-off by tracking which activities they enter, retain or leave, and how their invention and R D change alongside these choices.

Proposed contribution

We study how material constraints change firms' production choices, conditional on the technological knowledge they already have. For each activity a firm could enter, we measure two things: how much it depends on the scarce material, and whether the firm already has the knowledge needed to produce it. This two-dimensional map of opportunities helps explain when shortages trigger diversification and when firms remain in their existing activities.

Using firms' full patent portfolios, we also ask whether new inventions relate to activities the firm already produces or to activities it later enters. This lets us distinguish defensive, material-saving innovation from innovation that prepares a move.

Empirical approach

Around supply disruptions, we compare changes in production and patent portfolios across firms with different pre-shock material dependence, in an event-study / difference-in-differences spirit. Patent responses help interpret production changes. We are explicit about the limits: the disruptions identify the effect of material constraints on both outcomes, but they cannot by themselves establish that innovation causes diversification.

What you will get out of this project

? A frontier question at the intersection of industrial organisation, innovation economics and the economics of the energy transition, with direct relevance for industrial and trade policy.

? A full empirical pipeline: building a firm-level panel from heterogeneous sources, entity matching across patent and financial databases, and event-study designs around real-world shocks.

? Text-as-data skills: measuring technological similarity and relatedness from patent text and product descriptions, for example with embeddings, and validating these measures.

? Work on large, research-grade data (PATSTAT, PatentsView, Refinitiv) that is used in top academic and policy research.

Données du sujet :

Data, measures and sources

1. Material dependence and supply constraints

Pre-shock material dependence measures and a record of supply disruptions will be provided. Some data work will be needed to reshape and align them with firm-level data.

2. Production portfolios

We measure entry, retention and exit by activity, the range of activities, and revenue shares. Segment revenues come from Refinitiv Workspace (provided). They will be complemented with information from firms' public annual reports and product descriptions (open access).

3. Innovation portfolios

We measure new patent families, their distribution across technology fields, and entry into technologies new to the firm. We build three families of measures:

1. the patent portfolio itself;
2. cross-sectional similarity between patents;
3. deviation from the firm's own technological trajectory over time.

Sources are PATSTAT through TIP for global patent records and PatentsView for U.S. patent text. Applicants and subsidiaries will be matched to firms over time. Patents measure inventive activity, not adoption.

Modèles envisagés du sujet :

empirical quasi-experimental models, n

event-study / difference-in-differences design around material supply disruptions.. Firms are compared by pre-shock material dependence, with production and patent portfolio changes as outcomes.

Conceptually, it uses a two-dimensional opportunity map combining technological proximity and material exposure.

Auxiliary models measure patent similarity, relatedness, and text-as-data embeddings, plus entity matching across patent and financial databases.

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Notation du sujet :

Commentaires :